Varendra University
Department of Computer Science and Engineering
Lab Assignment
Course Title: Digital Image Processing Lab
Course Code: CSE 424

Title: Development of a Deep Learning-Based Writer Identification System Using Mid-Level Image Processing Techniques on Handwritten Image Datasets

CO3: Examine the performance midlevel image analysis techniques
PO4. Conduct investigations of complex engineering problems using research methods including research-based knowledge, design of experiments, analysis and interpretation of data, and synthesis of information to provide valid conclusions (WK8).
K8: Engagement with selected knowledge in the current research literature of the discipline, awareness of the power of critical thinking and creative approaches to evaluate emerging issues
Submission date: May 21, 2026

1. Objective

The objective of this project is to develop a Writer Identification System using handwritten pages collected from multiple individuals. Students will apply mid-level image processing techniques, data augmentation, and machine learning/deep learning models to identify the writer of a handwritten document automatically.

This project integrates concepts from:
• Digital Image Processing
• Pattern Recognition
• Machine Learning
• Deep Learning
• Computer Vision

2. Dataset Collection

Dataset Requirements:
• Collect at least 5 formally handwritten pages from each writer.
• Handwriting should be written on plain white paper using black or blue ink.
• Maintain consistent scanning or image capturing conditions.
• Images should be clear, properly aligned, and free from excessive shadows.

Recommended Collection Guidelines:
• Use mobile scanner applications without filtering.
• Save images in JPG or PNG format.

3. Mid-Level Image Processing

Students must select any 5 mid-level image processing techniques from the following list and apply them to all collected images.

Available Techniques:
• Laplacian Filtering
• Gabor Filtering

- Morphological Processing
- Skeletonization
- K-Means Segmentation
- Histogram Equalization
- Sharpening Filter
- Gaussian Blur
- Canny Edge Detector
- Sobel Operator

Processing Requirement:
If:
- Number of original images = 5
- Number of selected processing techniques = 5

Then:
- Generated processed images = 5 × 5 = 25 images

4. Data Augmentation

Students must further increase the dataset size by applying any 4 augmentation techniques to the processed images.

Available Augmentation Techniques:
- Random Rotation
- Random Horizontal Translation
- Gaussian Noise Injection
- Random Sharpness Adjustment
- Elastic Distortion

Augmentation Requirement:
If:
- Processed images = 25
- Selected augmentations = 4

Then:
- Final generated images = 25 × 4 = 100 images

5. Model Development

Students may select any suitable model for writer identification.

Option A: Machine Learning Models
- SVM
- XGBoost
- Random Forest
- k-NN

Option B: Pretrained CNN Models
- ResNet50
- EfficientNetV3
- InceptionV3
- GoogLeNet
- ConvNeXt

Option C: Transformer-Based Models
• Vision Transformer (ViT)
• Swin Transformer

Students are encouraged to compare multiple models if computational resources are available.

6. Report Writing Format

1. Introduction

Discuss the importance of handwriting analysis, significance of writer identification systems, real-world applications, and project motivation.

2. Materials and Methods

2.1 Dataset Collection

Describe:
• Number of writers
• Number of pages collected
• Data collection environment
• Image acquisition method
• Image format and resolution

2.2 Mid-Level Image Processing

Describe:
• Selected processing techniques
• Mathematical or conceptual overview
• Purpose of each technique
• Justification for selecting the techniques

Include:
• Original image
• Processed image examples
• Comparative visual analysis

2.3 Data Augmentation

Describe:
• Selected augmentation methods
• Purpose of each augmentation
• Effect on handwriting variation
• How augmentation improves model generalization

Include sample augmented images.

2.4 Model Architecture

Describe:
• Selected model
• Model architecture overview
• Input size
• Training parameters
• Optimizer
• Learning rate
• Batch size
• Number of epochs

2.5 Training and Testing Procedure

Describe:
- Dataset splitting strategy
- Hardware/software configuration
- Performance evaluation metrics

Required Evaluation Metrics:
- Accuracy
- Precision
- Recall
- F1-Score

Optional:
- Confusion Matrix
- ROC Curve
- Training/Validation Loss Graph

3. Results and Discussion

Students should:
- Present quantitative results
- Compare performance of selected techniques/models
- Discuss strengths and limitations

The discussion must include:
- Why selected mid-level techniques were effective
- Situations where performance improved
- Cases where the model failed
- Possible reasons for errors
- Future improvement opportunities

4. Conclusion

The conclusion should summarize:
- Project objective
- Methods used
- Key findings
- Overall performance
- Future research directions

The conclusion should be formal, concise, and academically structured.

5. Acknowledgement (If any)

6. References

- IEEE Style

7. Submission Requirements

Students must submit:
- Final report
- Source code

8. Evaluation Criteria

- Dataset Collection – 20%
- Mid-Level Processing – 35%

- Augmentation Strategy – 5%
- Model Development – 20%
- Report Quality – 20%

9. Important Academic Notes

- Direct copying from online sources is prohibited.
- Proper citation must be maintained.
- Students are encouraged to perform comparative analysis.
- Clear explanation of image processing techniques is mandatory.
- Proper figure captions and tables must be included in the report.